

AIR Lab Website Redesign

Complete Strategic & Full-Stack Implementation Plan

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Target: Independent AIR Lab Portal

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1 Executive Summary & The "Why"

Why Rebrand as an Independent Site?

The current site (`cct.neduet.edu.pk/AIR`) is an outdated subpage under the departmental directory. Top global labs (e.g., **Stanford AI Lab** `ai.stanford.edu`, **Allen AI** `allenai.org`) operate independent portals. A dedicated web presence establishes authority for international grants, corporate industry partnerships, and researcher recruitment.

Why the MERN Stack?

University faculty require a dynamic **Admin Dashboard** where faculty and RAs can update publications, add graduating student batches, and announce research vacancies without writing code or filing tickets with university network administrators.

Core Architectural Objective: Build a high-performance single-page application (React + Vite + Tailwind CSS) powered by a lightweight Node.js/Express REST API and MongoDB Atlas. It features a decoupled design: if the backend or database ever undergoes maintenance, the public frontend seamlessly serves cached data without disruption.

2 4-Week Milestone Roadmap (Post-Midterms Schedule)

Phase & Window	Primary Milestone	Key Deliverables & Action Items	Review Gate
Week 1 CURRENT	Branding & Data Blueprint	- Finalize design system: Inter + Geist Mono fonts, light/dark tokens. - Set up Google Sheets pipeline for member bios & photos. - Collect FYDP group abstracts (5-6 groups under supervision). - Lock MongoDB JSON schemas for all 6 collections.	Supervisor sign-off on sitemap & color tokens.
Week 2	Frontend Prototype Complete	- Interactive React + Vite application ready with mock data. - Animated neural canvas hero, statistics strip, & bento grid. - Honeycomb team view, 4-tier project tabs, publication list. - Live interactive Theme Switcher (Light & Dark modes).	Design freeze: Supervisor tests and approves live UI.
 MIDTERM EXAMINATIONS RECESS — ACADEMIC PAUSE			
Week 3 POST-MIDS	MERN Backend & Database	- Provision MongoDB Atlas production cluster. - Build Node.js + Express REST API with input validation. - Connect Cloudinary / S3 CDN for compressed image storage. - Implement JWT-based secure authentication for admins.	Postman test suite passing; live data feeds into frontend.
Week 4 POST-MIDS	Vacancy Engine & Handover	- Deploy live Vacancy Indicator (Open/Closed slots). - Build secure Admin Panel for non-technical content management. - Cross-device optimization (mobile, tablet, desktop). - Domain mapping (<code>air.neduet.edu.pk</code>) & admin handover guide.	Full deployment, final demo, and documentation handover.

3 Detailed Feature Architecture & Technical Specifications

A. The Vacancy Indicator Feature

Problem: Faculty receive dozens of redundant emails weekly asking about research assistantships, FYDP supervision openings, and internships.

Solution: A real-time vacancy module controlled via the Admin Panel that displays live badges across the homepage and careers page:

- **OPEN Active Vacancies:** Displays available slots (e.g., "2 RAs for Regional LLM Project", "1 Undergraduate FYDP Group") with application deadline and direct submission link.
- **REVIEWING Reviewing Applications:** Submissions closed; committee evaluating candidates.
- **CLOSED No Vacancies:** Displays notice with date for the next recruitment cycle.

B. Content Architecture & Required Collections

Section	Category Breakdown	Technical Fields / Schema
Team Roster	(i) Faculty Members (ii) Postgraduate Students (iii) Undergraduate Scholars	name, role, category, bio, image_url, email, google_scholar, linkedin, order
Projects	(i) Funded Projects (ii) R&D Projects (iii) Undergraduate (FYDP) (iv) Postgraduate Dissertations	title, category, abstract, pi_supervisor, funding_agency, grant_amount, demo_url, status, metric_highlight
Publications & Blogs	(i) Peer-Reviewed Research Papers (ii) Lab Articles & Updates	title, authors, venue, year, doi_link, pdf_url, blog_content, publish_date, cover_image
Lab Gallery	Event photos, symposiums, lab equipment, hardware prototypes	title, caption, image_url, date, span_class
Vacancies	Research Assistants, Interns, FYDP Slots	role, category, slots, status, deadline, apply_url

4 Web Dev Industry Consensus (Mitigating Academic Lab Pitfalls)

Real-World Risk	Why Most Student Sites Fail	Our Strategic Mitigation
1. The "Graduation Trap"	When developers graduate, unmaintained custom servers crash or database credentials expire, causing the website to go dark.	Decoupled JSON Fallback: The React app builds static cache fallbacks. If the Node/MongoDB server goes down, the frontend still renders complete lab data gracefully.
2. Admin Over-Engineering	Teams try building a complex drag-and-drop CMS from scratch, run out of time, and leave half-working forms.	Focused CRUD Forms: Pre-built, robust table and modal components strictly tailored to the 5 core models, keeping maintenance zero-friction.
3. Image Storage Bloat	Storing raw 8MB event photos in the database or server disk slows page loads and consumes disk quota.	Automated Cloud CDN: Integrate Cloudinary free tier with automatic web compression and auto-format (f_auto, q_auto).

5 Agenda for Tomorrow's Meeting with the Supervisor

1. **Show Working Prototype Live:** Open `http://localhost:5173/AIR-proto/` on your laptop. Show the interactive neural canvas, the honeycomb team layout, and demonstrate switching between Light and Dark modes.
2. **Clarify the Color Strategy:** Explain that instead of a literal rainbow palette, we engineered "*Chromatic / Spectral AI Accents*"—a prestigious Stanford-style neutral base with vibrant jewel-tone accent chips for each research pillar.
3. **Walk Through the Roadmap:** Present the 4-Week Post-Mids plan to demonstrate thorough planning and realistic workload management.
4. **Lock In These 4 Decisions:**
 - **Domain:** Will NED provide `air.neduet.edu.pk` or should we configure an independent domain?
 - **FYDP Data:** Can Hasan circulate a standardized Google Form to the 5 supervised groups to collect project summaries?
 - **Vacancy Intake:** Should the vacancy button link to a Google Form or a built-in upload form?
 - **Sign-off:** Confirm the Team (3 tabs) and Project (4 tabs) categories are frozen.